University of Nottingham
UK | CHINA | MALAYSIA

Department of Computer Science

Abstract

Problem: DRL navigation policies achieve >85% success in open environments but catastrophically fail (0% SR) in trap scenarios requiring temporary retreat from the goal. **Gap:** Exploration rewards fix traps (0→83%) but destroy precision (narrow door 100→0%). Reward annealing cannot resolve this conflict. **Method:** We propose Stall-Triggered Policy Switching (STPS), a training-free runtime architecture that detects motion stagnation and dynamically switches between precision and exploration policies. **Result:** STPS achieves 94% average scenario SR (+27 points vs baseline), is the only method without catastrophic failure on any scenario, and matches baseline on standard environments (88%).

Introduction

Deep reinforcement learning (DRL) has become standard for mapless robot navigation, mapping raw LiDAR to velocity commands. CNN-TD3 achieves high success rates in cluttered environments.

However, in **trap scenarios** — U-shaped dead ends, narrow corridors requiring retreat — greedy goal-directed rewards cause the robot to oscillate until timeout.

Key Finding: Exploration–Precision Conflict

Exploration bonus solves U-trap (0→83%) but **kills narrow door (100→0%).**

Annealing recovers narrow door but loses U-trap.

Method	Std	U-trap	Narrow	Avg
Baseline	87%	0%	100%	67%
Curriculum	74%	0%	8%	52%
Exploration	80%	83%	0%	71%
Annealed	82%	0%	100%	75%

Method: Stall-Triggered Policy Switching (STPS)

MAIN MODE
Precision policy πP
Standard goal-directed navigation (DEFAULT)

Stall: $\|x_t - x_{t-20}\| < 0.15$m OR oscillation≥5/12 →

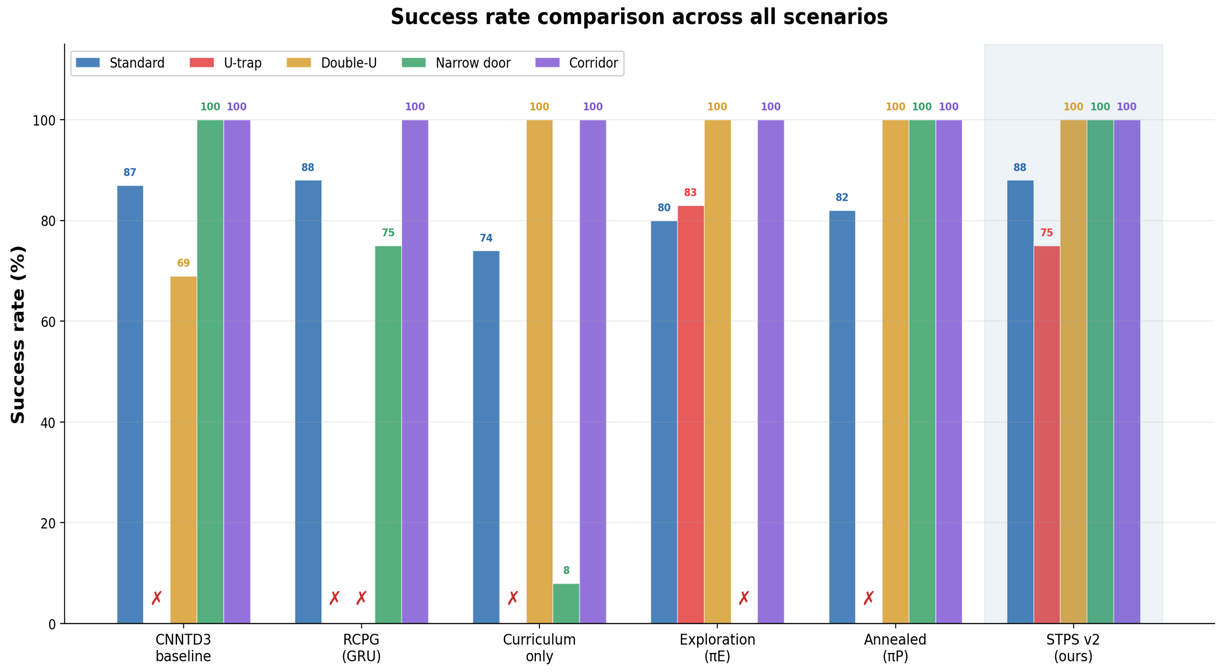
← Escaped: $k \geq 120$ AND $\|x_t - x_{sw}\| > 0.5$m

ESCAPE MODE
Exploration policy πE
Novelty-seeking navigation (TEMPORARY)

- Zero training cost: Frozen policies + runtime logic
- <0.1ms overhead: One distance check per step
- Real-robot ready: Simple if-else, deployable on LIMO

Main Results: STPS Outperforms All Single Policies

Method	Standard	U-trap	Double-U	Narrow	Corridor	Scenario Avg
CNNTD3 baseline	87%	0±0%	69±4%	100±0%	100±0%	67%
RCPG (GRU)	88%	0±0%	0±0%	75±0%	100±0%	44%
Curriculum only	74%	0±0%	100±0%	8±0%	100±0%	52%
Exploration (πE)	80%	83±0%	100±0%	0±0%	100±0%	71%
Annealed (πP)	82%	0±0%	100±0%	100±0%	100±0%	75%
STPS v2 (ours)	**88%**	**75±7%**	**100±0%**	**100±0%**	**100±0%**	**94%**



Training Curves

Parameter Sensitivity

	δ=0.10	δ=0.15	δ=0.20
W=15	42/100	42/100	67/100
W=20	**67/100**	**67/100**	**67/100**
W=30	0/100	17/100	25/100

Narrow door = 100% across entire grid → switching never harms precision

W=20: stable 67% U-trap regardless of δ → no per-env tuning needed

What didn't work:

- STPS v3 (progressive escape): no gain
- U-trap specialist model: 83% = original improved
- Bottleneck = geometric difficulty

Conclusion

1. Exploration rewards create a hard, irreconcilable conflict with precision navigation in DRL.
2. STPS resolves this by composing complementary frozen policies at runtime with stall/oscillation detection.
3. The method requires zero additional training and adds negligible (<0.1ms) computational overhead.

Application

Sim-to-real advantages:

- STPS switching logic is pure Python if-else — no GPU, no retrain
- Both policies use standard LiDAR input, directly transferable

Deployable scenarios:

- Warehouse navigation with shelf dead-ends
- Any indoor space with concave obstacles

Jiabao Cao,Shengqin Jiang,Wenjing Chen,Zexian Xiong,Zhirui Shen
Supervisor: Fuhua Jia ,Tianxiang Cui